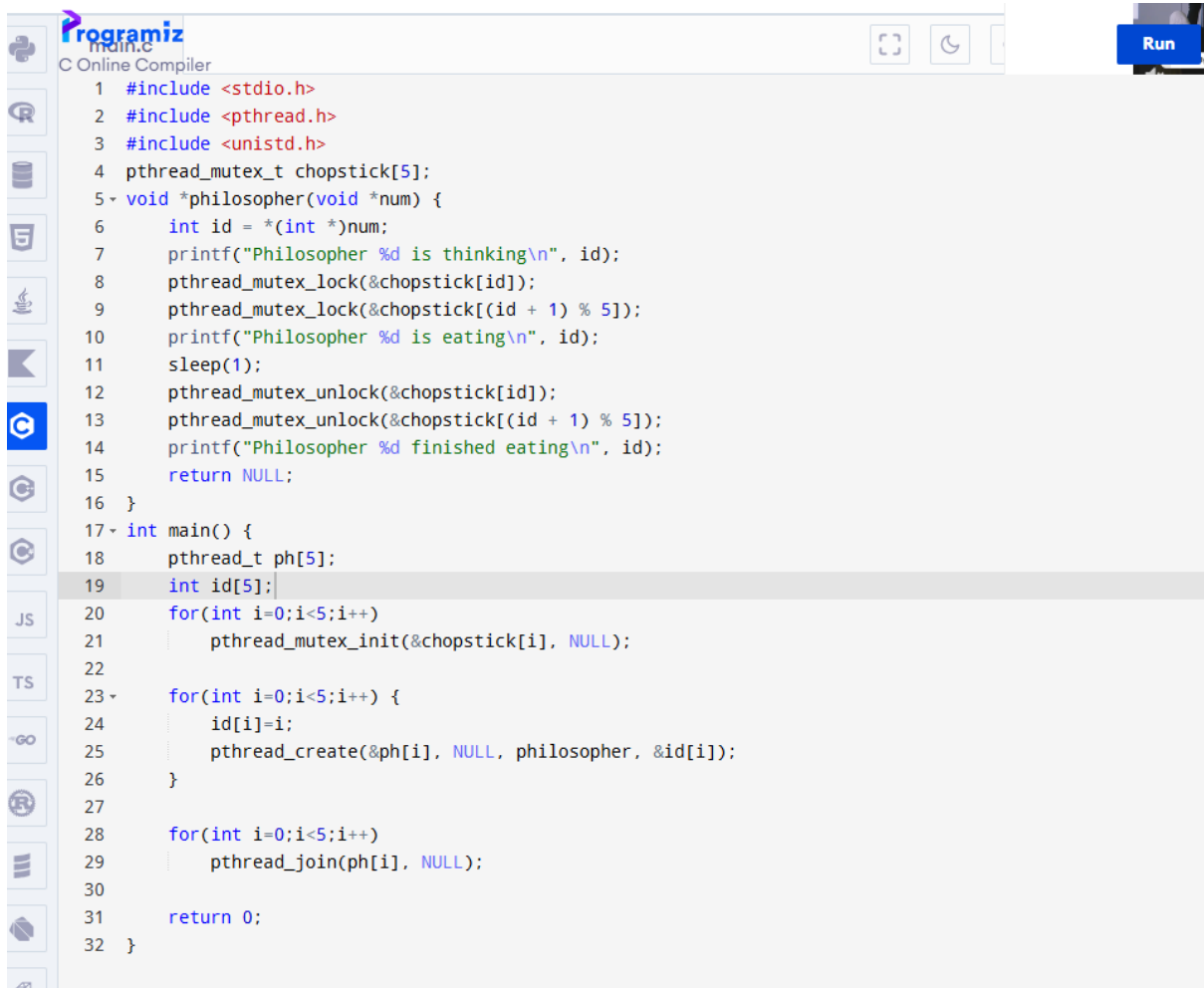


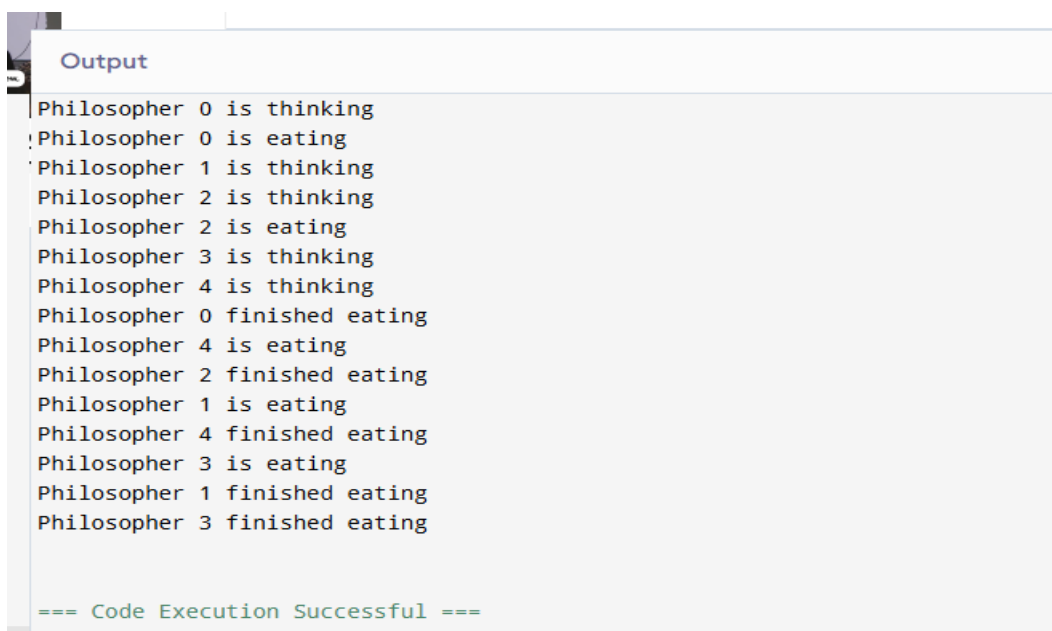
12. Design a C program to simulate the concept of Dining-Philosophers problem

Code:



```
1 #include <stdio.h>
2 #include <pthread.h>
3 #include <unistd.h>
4 pthread_mutex_t chopstick[5];
5 void *philosopher(void *num) {
6     int id = *(int *)num;
7     printf("Philosopher %d is thinking\n", id);
8     pthread_mutex_lock(&chopstick[id]);
9     pthread_mutex_lock(&chopstick[(id + 1) % 5]);
10    printf("Philosopher %d is eating\n", id);
11    sleep(1);
12    pthread_mutex_unlock(&chopstick[id]);
13    pthread_mutex_unlock(&chopstick[(id + 1) % 5]);
14    printf("Philosopher %d finished eating\n", id);
15    return NULL;
16 }
17 int main() {
18     pthread_t ph[5];
19     int id[5];
20     for(int i=0;i<5;i++)
21         pthread_mutex_init(&chopstick[i], NULL);
22
23     for(int i=0;i<5;i++) {
24         id[i]=i;
25         pthread_create(&ph[i], NULL, philosopher, &id[i]);
26     }
27
28     for(int i=0;i<5;i++)
29         pthread_join(ph[i], NULL);
30
31     return 0;
32 }
```

Output:



```
Philosopher 0 is thinking
Philosopher 0 is eating
Philosopher 1 is thinking
Philosopher 2 is thinking
Philosopher 2 is eating
Philosopher 3 is thinking
Philosopher 4 is thinking
Philosopher 0 finished eating
Philosopher 4 is eating
Philosopher 2 finished eating
Philosopher 1 is eating
Philosopher 4 finished eating
Philosopher 3 is eating
Philosopher 1 finished eating
Philosopher 3 finished eating

=== Code Execution Successful ===
```

